

A Comparative Analysis of KNN Regression and Random Forest for Uber and Lyft Price Prediction

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Abstract—Ride-sharing platforms such as Uber and Lyft use dynamic pricing that makes fare prediction a challenging problem. This study compares K-Nearest Neighbors (KNN) Regression and Random Forest in predicting ride-sharing prices using a dataset of 637,976 ride records from Boston, MA. Features include distance, surge multiplier, ride type, source, destination, and time-based attributes. KNN ($k=9$) achieved an RMSE of \$1.841, MAE of \$1.187, and R^2 of 0.961, while Random Forest achieved an RMSE of \$1.811, MAE of \$1.142, and R^2 of 0.962. Results show that both models effectively predict ride-sharing prices, with Random Forest outperforming KNN overall.

Index Terms—ride-sharing, price prediction, KNN regression, random forest, machine learning, Uber, Lyft, dynamic pricing

I. INTRODUCTION

Ride-sharing platforms such as Uber and Lyft have become a common mode of transportation, yet their dynamic pricing models make it difficult for consumers to anticipate fares. Predicting these prices accurately has practical value for travelers seeking to budget their transportation costs.

This study addresses the following research question: *To what extent do KNN Regression and Random Forest differ in their ability to predict ride-sharing prices across different times of day and locations?* Using a dataset of 637,976 ride records from Boston, MA, both models were trained on features including distance, surge multiplier, ride type, and time-based attributes. KNN achieved an R^2 of 0.961 and Random Forest achieved an R^2 of 0.962, with Random Forest outperforming KNN as the better overall model.

II. PROBLEM DESCRIPTION

A. Dataset

Our data source came directly from the project's canvas page, originally sourced from Kaggle [1]. We chose the Uber and Lyft Cab Prices data set. The dataset contains 637,976 ride records collected in Boston, MA.

B. Input features

The categorical features were `cab_type`, `name`, `source`, and `destination` were converted to numerical representations using a hot encoding by using the command `pd.get_dummies`. The numerical features used in the models include `distance`, `hour`, `day_of_week`, and `surge_multiplier`. The features `hour` and `day_of_week` were engineered from the raw `time_stamp` column.

The columns `id` and `product_id` were dropped as they provide no predictive value.

C. Target output

The target variable is `price`, a continuous numerical value representing the ride fare in USD. This is a supervised regression task.

D. Preprocessing step

The following preprocessing steps were applied to prepare the dataset for modeling. First, the irrelevant columns `id`, `product_id`, and `time_stamp` were dropped. Next, `hour` and `day_of_week` were engineered from the raw timestamp. All categorical variables were then converted using one-hot encoding, and rows with missing values were removed. Finally, `StandardScaler()` was applied to the KNN training features only, as KNN is sensitive to feature magnitude while Random Forest is not.

III. MODELS USED

A. K-Nearest Neighbors Regression

Our first choice of model is K-Nearest Neighbor [2]. KNN is a non-parametric algorithm that predicts a target value by finding the k most similar training observations and averaging their target values. Due to the large dataset's size of 637,976 rows, KNN was trained on a random sample of 200,000 training observations to manage computational constraints. Feature via `StandardScaler()` was applied, as KNN is sensitive to differences in feature magnitude. The optimal k was determined by evaluating odd values $k \in \{1, 3, 5, 7, 9, 11\}$, where the optimal value is $k = 9$.

B. Random Forest Regression

Our second model is Random Forest Regression [3]. Random Forest is an ensemble learning method that constructs multiple decision trees during training and outputs the average prediction across all trees. This approach reduces overfitting and handles non-linear relationships and feature interactions effectively. Unlike KNN, Random Forest does not require feature scaling and was trained on the full training set of approximately 510,000 observations. An additional advantage of Random Forest is its ability to provide feature importance rankings, which offer insight into which variables most strongly influence ride-sharing prices.

IV. PERFORMANCE INDICATORS AND MODEL COMPARISON

A. Evaluation metrics

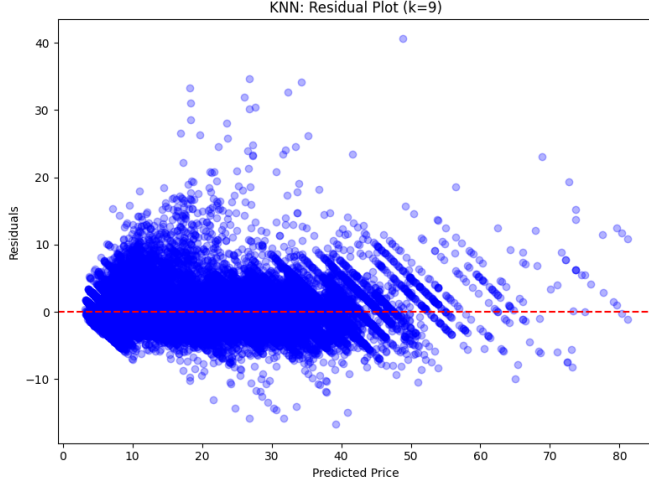


Fig. 1. KNN Residual Plot

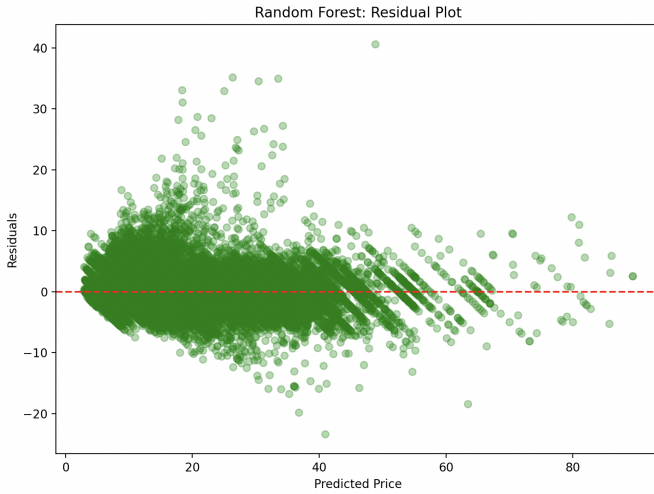


Fig. 2. Random Forest Residual Plot

Figure 1 and 2 show that most residuals centered around zero, indicating strong overall performance. However a fan-shaped pattern at higher prediction price suggests the model becomes less accurate for more expensive rides. Vertical banding reflects the discrete nature of one-hot encoded features, and a few outliers above \$20 indicate the model struggles with premium ride types and high surge multiplier scenarios.

TABLE I
FEATURE IMPORTANCE TABLE

	Feature	Importance
10	name_Lux Black SUV	0.252380
7	name_Black SUV	0.252263
0	distance	0.154810
9	name_Lux Black	0.092021
6	name_Black	0.071042
1	surge_multiplier	0.041319
8	name_Lux	0.040658
12	name_Lyft XL	0.033073
17	name_UberXL	0.029048
2	hour	0.009776
⋮	⋮	⋮

For our Random Forest Regressor we choose to show the Table I. This shows how important each input is and how it affects the overall price of the ride. The results show that ride type features, particularly Lux Black XL and Black SUV, are the most important predictors, meaning that premium vehicle options significantly increase price. Distance is also a key factor, which is expected as longer trips generally lead to higher fares.

B. Confidence Intervals

The 95% confidence interval for the Random Forest's model RMSE is $[1.786, 1.838]$, while the KNN's model falls within $[1.817, 1.875]$. This means that across different samples of test data, the prediction error of Random Forest is consistently lower than that of KNN. Additionally, the Random Forest interval is slight narrower, meaning that its performance is more stable and reliable, whereas KNN shows slightly greater variability. These results suggest that the Random Forest model not only achieves better accuracy but also provides more consistent predictions compared to KNN.

C. Results

The performance of the KNN and Random Forest models was evaluated using RMSE, MAE, and R^2 . The results are summarized in table II

TABLE II
PERFORMANCE COMPARISON OF KNN AND RANDOM FOREST

Model	RMSE	MAE	R^2
KNN (k=9)	1.841	1.187	0.961
Random Forest	1.811	1.142	0.962

The Random Forest model slightly outperforms the KNN model across all evaluation metrics. It achieves a lower RMSE and MAE, and a lightly higher R^2 , suggesting more accurate predictions and better overall fit to the data. A manual prediction was conducted to further evaluate the models on a specific example:

- Actual Price: \$16.50
- KNN Prediction: \$17.50
- Random Forest Prediction: \$17.02

Both models overestimated the actual price; however, the Random Forest prediction is closer to true value, with \$0.52 error, compared to \$1.00 for KNN.

D. Best Model Justification

While both models are well trained, we have to choose just one to be the best. Based on the evaluation metrics and confidence interval, the Random Forest model is selected as the best-performing model. It achieves lower RMSE and MAE, along with a slightly higher R^2 compared to KNN. Additionally, its narrower confidence interval demonstrate more stable and reliable performance across different samples. Therefore, Random Forest is more effective for capturing the underlying patterns in ride pricing.

V. PROS AND CONS

A. K-Nearest Neighbors Regression

KNN offers several advantages that make it a suitable model for ride-sharing price prediction. KNN is a non-parametric model that makes no assumptions about the underlying data distribution, which is beneficial given the dynamic nature of ride-sharing pricing. It is also simple and intuitive model that requires no assumptions about the data. It predicts ride prices by finding the most similar rides in the training set and averaging their fares, making it easier to understand.

Despite its strengths, KNN Regression has several notable limitations. The most significant drawback is its computational cost, with a dataset of 637,976 rows, the model was too slow to run on the full data and had to limited to 200,000 rows, which may have caused it to miss some patterns. Having 43 features also hurt KNN's performance, as too many features make it harder for the model to truly find similar rides.

B. Random Forest Regressor

Random Forest handles all the input features to predict the final price; it shows hidden or unseeable feature relations to the price, and it works well with large datasets. It was trained on the full training set without requiring sampling and provides feature importance rankings that offer meaningful insight into which variables drive ride-sharing prices.

However, with all great things comes cons. Random Forest has a high computation time, a bit harder to interpret at the individual tree level. Additionally, while using all the features to predict the final price, it can occasionally assign weight to less relevant features, potentially introducing noise into predictions.

C. Overall Approach

Using two contrasting models enables a meaningful comparison of complexity vs. simplicity. Evaluating both on the same test set ensure fairness. Confidence intervals add statistical rigor beyond point estimates.

A key limitation to this approach is that KNN required sampling to 200,000 rows due to computational constraints, which may disadvantage it in the comparison against Random Forest, which was trained on the full dataset. Additionally, the

dataset is limited to the Boston, MA area, which may limit the generalizability of these findings to other cities or regions.

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